SILTHAUL · USER REQUIREMENTS

SiltHaul user requirements

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
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REVISION HISTORY

Version	Date	Author	Change
0.1	2026-11-03	Amish Chadha	First draft for the kit 1.9.1 upgrade, from the users and stakeholders in SLH-PRB-001 and the twelve requirements Amish accepted in SLH-DDR-003 and SLH-DDR-004; to review with a co-design partner

SiltHaul user requirements

What users need, written from their side. The thirteen needs below were drafted from the users and stakeholders in SLH-PRB-001 (Tables 1 and 2), used to score the concept options (SLH-OPT-001), and then traced to the product requirements in SLH-PRS-001 (SLH-RTM-001). The twelve original requirements R1 to R12 that Amish accepted on 2026-10-03 are kept as product requirements PR-001 to PR-012; nothing Amish decided was changed. The needs remain a desk draft until a co-design partner reviews them with the people who clear flood mud. Method: STANDARDS section 22.

Table 1. User requirements.

ID	Category	Title	Need	Priority	Source
UR-001	Performance	Nobody lifts or carries mud	The mud goes out of the room without anyone lifting it or carrying it	High	Householders and volunteers (PRB Table 1); Amish's restatement of R1, SLH-DDR-003
UR-002	Performance	Keeps up with a bucket crew	A crew of four moves at least half as much mud an hour as the same crew with shovels and buckets	High	Householders and volunteers; relief NGOs
UR-003	Usability	Ordinary volunteers can work it	A person of ordinary strength can work it after a short briefing, with no heavy single effort	High	Relief NGOs (PRB Table 1)
UR-004	Deployment and assembly	Gets through our door	It works through a doorway about 0.7 m wide	High	Householders; PRB operating environment
UR-005	Performance	Reaches the far end of the room	It clears a room up to 15 m from where the crew works	Medium	Municipal drain and sanitation crews
UR-006	Safety	Safe near ropes, anchors and moving parts	Nobody is hurt by a rope that lets go, an anchor that fails, a load that runs back or a moving part	High	Relief NGOs; disaster authorities (PRB Table 2)
UR-007	Deployment and assembly	Quick to set up	Three untrained volunteers set it up in 20 minutes after a short briefing	Medium	Relief NGOs
UR-008	Logistics	Travels in a small car	The kit fits a small hatchback and no single part is more than 25 kg	Medium	Relief NGOs; PRB constraints
UR-009	Cost	Affordable for a relief group	The prototype parts stay at or under the USD 2,000 value-engineering target	Medium	Relief NGOs; PRB constraints
UR-010	Maintenance	Made and repaired locally	A local fabricator can make and mend it from stock steel, rope and commercial parts with hand tools and a small welder	High	Local fabricators (PRB Table 1)
UR-011	Environment and durability	Survives mud and wet	It keeps working in wet mud and can be washed down	Medium	All users; PRB operating environment
UR-012	Safety	Keeps people out of the mud	The crew spends less time standing in and touching contaminated mud, and the crank crew stays clear of it	High	Householders and volunteers; local health authorities
UR-013	Environment and durability	Adds no wet waste	It moves the mud as it is, with no added water to dispose of	Low	Municipal drain and sanitation crews